PPE COMPLIANCE DETECTOR

Automated PPE Verification for Home Health Caregivers

ITAI 1378 — Final Presentation

MODEL TRAINED ✓ 90% mAP

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PROJECT TIER

Tier 1

The Problem

- Home health caregivers are required to wear PPE (masks, gloves) during certain visits.
- Compliance today relies on self-reporting and occasional manual supervisor spot-checks.
- This does not scale across a large caregiver workforce and creates gaps during infection-control audits.

Who cares: Agency compliance officers, supervisors, and payer/UHC infection-control audits.

MANUAL CHECKS DON'T SCALE

1

spot-check per visit — no continuous verification

Why it matters

Missed PPE compliance is both a health risk and a documentation liability during audits.

Our Solution

An object detection system that scans a photo of a caregiver and automatically flags PPE compliance — in real time, with no manual review.

1

Capture

A photo of the caregiver is taken at check-in or during a visit.

→

2

Detect

A fine-tuned YOLOv8 model scans the image for mask presence and placement.

→

3

Flag

The system outputs a clear compliant / non-compliant result instantly.

Compliant → mask detected correctly ⚠ **Non-Compliant** → mask missing or worn incorrectly

Technical Approach

TECHNIQUE	MODEL	FRAMEWORK
Object Detection	**YOLOv8n**	**Ultralytics (PyTorch)**
Bounding boxes localize exactly where PPE is present or missing — not just a whole-image label.	Fine-tuned for 75 epochs on free Colab GPU — fast to train, strong results for its size.	Mature tooling, strong community support for PPE-style datasets, minimal boilerplate.

Why this stack: kept scope realistic for the timeline — real training and evaluation, without custom architecture design or manual labeling.

Data Plan

DATASET SIZE

4,547

labeled images

Source

Roboflow Universe
“Mask-Detection-YOLOv8” (AGH)

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Labels (3 classes)

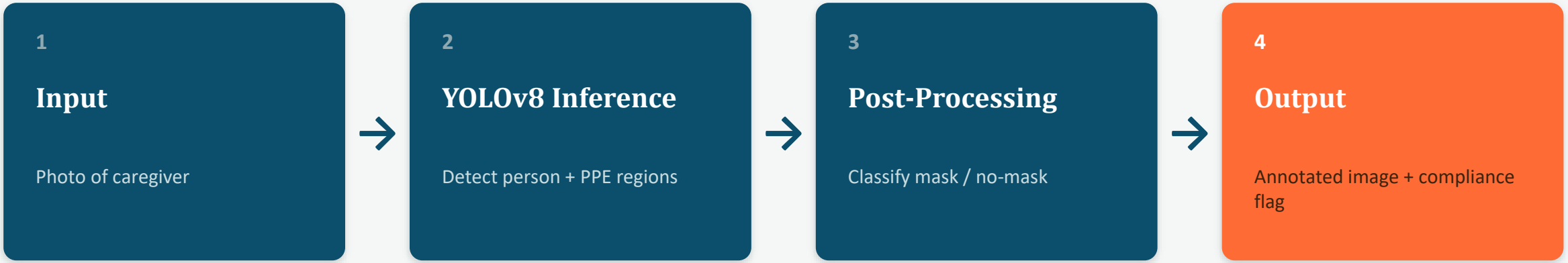
- with_mask
- without_mask
- incorrectly_worn_mask

Core deliverable: with_mask / without_mask. Incorrect-mask detection is a stretch goal.

Preparation

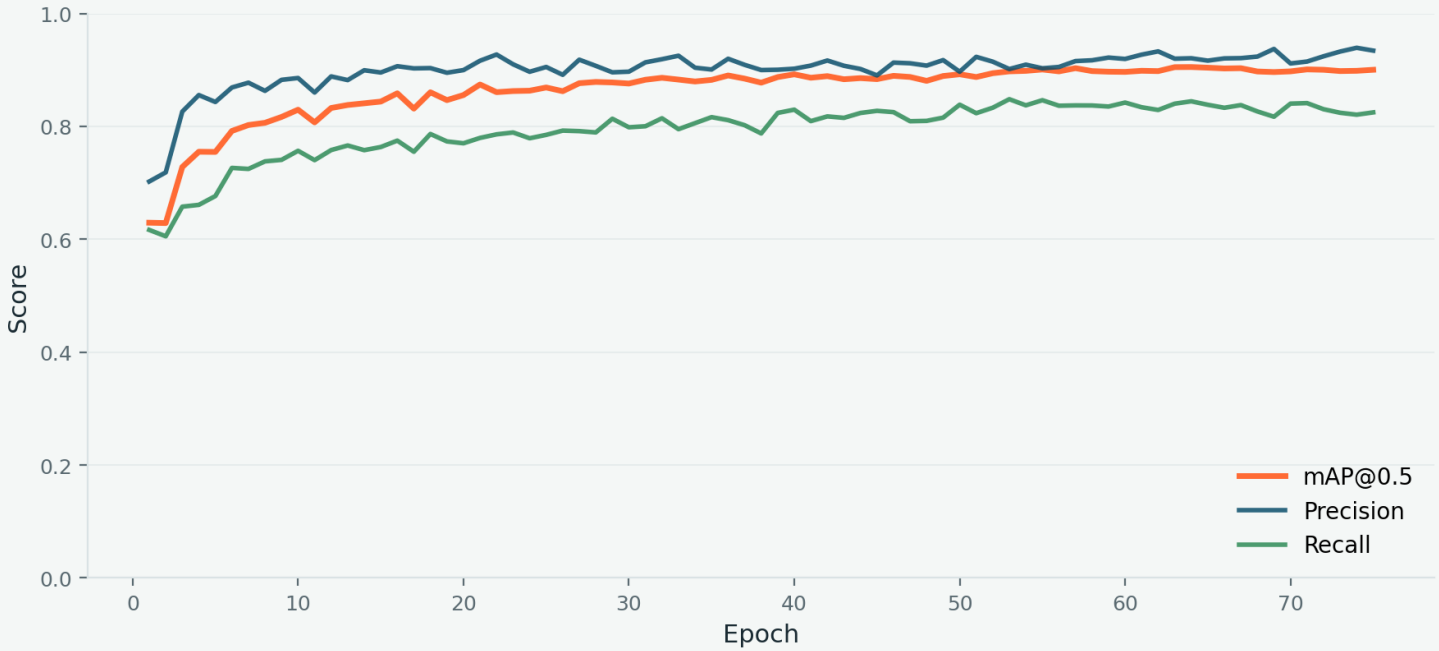
- No manual labeling — dataset ships pre-annotated in YOLO format.
- Exported directly via the Roboflow Python package.
- Standard Ultralytics augmentation applied at train time (flip, mosaic, brightness).

System Diagram



Pipeline is intentionally linear — one photo in, one clear compliance flag out. No manual review step required.

Training Results



mAP@0.5
90.0%

Precision
93.4%

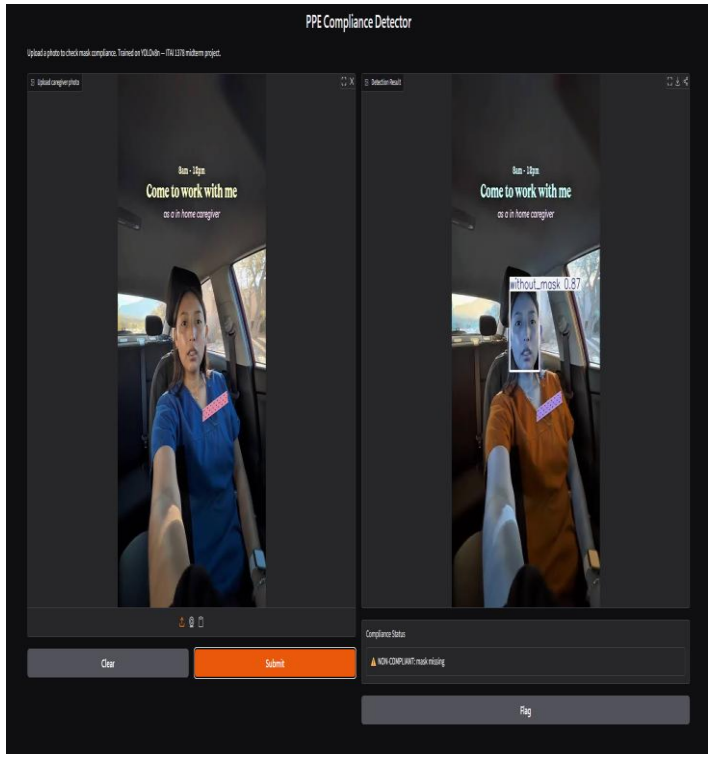
Recall
82.5%

Model exceeded our 75–80% mAP target, matching published benchmarks (~91%) for similar fine-tuned models on this dataset.

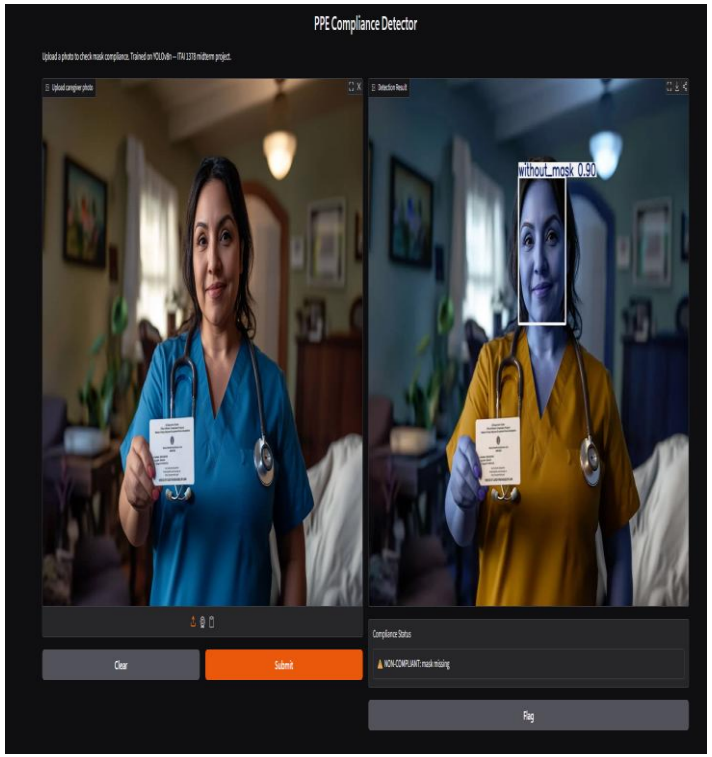
75 epochs · 76.4 min total training time on Colab T4 GPU

Live Demo

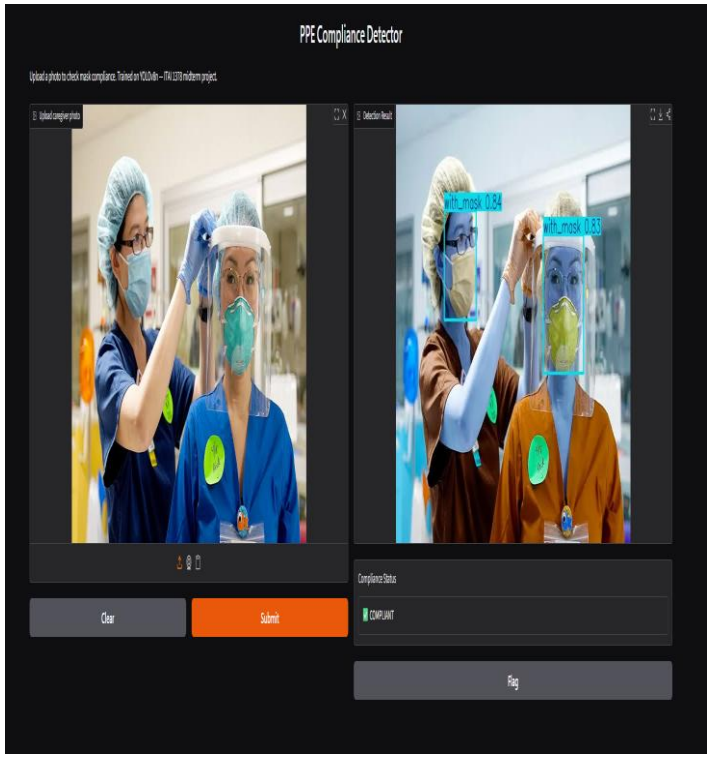
Interactive Gradio interface — upload a photo, get an instant compliance flag.



NON-COMPLIANT — mask missing (0.87)



NON-COMPLIANT — mask missing (0.90)



COMPLIANT — both masks detected (0.84, 0.83)

Success Metrics: Target vs. Actual

PRIMARY METRIC — mAP@0.5

Target: ≥ 75–80%

90.0%

✓ TARGET EXCEEDED

Best single checkpoint (epoch 64) hit 90.5% — model stayed stable through training.

SECONDARY METRIC — SPEED

Target: < 0.5s / image

Precision 93.4%

Recall 82.5%

Confirmed fast, responsive inference during live Gradio demo — well under a second per image.

Project Timeline

✓	✓	✓	✓	✓	✓
Dataset + environment	**Sanity-check run**	**Full training + tuning**	**Build demo**	**Docs, slides, rehearse**	**Present**
Complete	*Complete*	*90% mAP reached*	*Gradio demo live*	*Complete*	🎉 *Today*

Challenges & What We Learned

Challenge	How We Handled It
Full 75-epoch run took ~76 min	Used checkpointing (save_period=10) so progress was never lost to a Colab disconnect.
Domain mismatch (dataset ≠ real home visits)	Accepted as a known limitation — can't use real patient photos for privacy reasons.
Confirming target vs. actual model quality	Cross-checked our 90% mAP against a published benchmark on the same dataset (~91%) to validate the result was reasonable, not a fluke.
Turning raw detections into a usable signal	Built a simple compliance-flag function on top of raw YOLO output, then wrapped it in a Gradio UI for a real interactive demo.

Conclusion

COMPUTE USED

Colab T4 GPU

76.4 min total training

FRAMEWORKS

Ultralytics YOLOv8

PyTorch · Roboflow · Gradio

TOTAL COST

$0

Free tiers covered the full project

A working PPE compliance detector — 90% mAP, real-time demo, built on public data and free compute.

Thank you — questions welcome